

6. In each of the following questions, select the correct answer. (2 pts. each)

- (a) Let A be a 3×3 matrix, such that $AX = -X$, $AY = Y$, $AZ = -2Z$, for nonzero vectors X, Y and Z in $\mathbb{R}^3$. Then
- (A) $\det(A) = 2$
 - (B) the characteristic equation of A is: $\lambda^3 + 2\lambda^2 - \lambda - 2 = 0$
 - (C) all of the above
 - (D) none of the above
- (b) Let A be a nonzero 4×7 matrix, then A may be
- (A) of rank 5
 - (B) of nullity 7
 - (C) row equivalent to a 7×4 matrix
 - (D) none of the above
- (c) Let A be an $n \times n$ matrix in a reduced row echelon form. Then
- (A) $A = I_n$
 - (B) $A^2 = A$
 - (C) all of the above
 - (D) none of the above
- (d) For all vectors U, V and W in $\mathbb{R}^n$
- (A) $\|U + V + W\| \leq \|U\| + \|V\| + \|W\|$
 - (B) $\|U + V + W\| = \|U\| + \|V\| + \|W\|$
 - (C) $\|U + V + W\| \geq \|U\| + \|V\| + \|W\|$
 - (D) none of the above
- (e) Let V be a vector space and W_1 and W_2 be subspaces of V . If $B_1 = \{X, Y\}$ and $B_2 = \{X, Z\}$ are bases for W_1 and W_2 , respectively, then
- (A) $\dim(V) \geq 4$
 - (B) $\dim(W_1 \cap W_2) = 1$
 - (C) all of the above
 - (D) none of the above

4. (a) $A = X^{-1}BX \implies A^3 = (X^{-1}BX)^3 = X^{-1}\underline{BXX^{-1}BXX^{-1}BX} = X^{-1}B^3X \implies A^3 \text{ and } B^3 \text{ are similar.}$

$$(b) \quad |\lambda I_3 - A| = \begin{vmatrix} \lambda + 1 & 0 & 0 \\ 0 & \lambda - 2 & 1 \\ -1 & 0 & \lambda - 2 \end{vmatrix} = (\lambda + 1)(\lambda - 2)^2.$$

Let $\lambda_1 = \lambda_2 = 2$, $\lambda_3 = -1$ be the eigenvalues of A .

$$\text{Solve: } (2I_3 - A)X = O, \quad \left[\begin{array}{ccc|c} 3 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 \end{array} \right] \equiv \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

The eigenvectors corresponding to the eigenvalues $\lambda_1 = \lambda_2 = 2$, are $X = r \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$,
 $r \neq 0$.

The algebraic multiplicity, of $\lambda = 2$ is 2, But the geometric multiplicity is 1.
 Therefore, A is not diagonalizable.

5. (a) Since X and Y are orthogonal, then $X \cdot Y = 0$. $\|X + Y\|^2 = (X + Y) \cdot (X + Y) = X \cdot X + 2X \cdot Y + Y \cdot Y = \|X\|^2 + 0 + \|Y\|^2 = \|X\|^2 + \|Y\|^2$.

- (b) (i) Consider $P_1(-1, 1, 2) \in L$. Let $V = \overrightarrow{P_1 P} = (2, 1, -3)$. Thus,

$$N = |(2, 1, -3) \times (2, -1, 1)| = \begin{vmatrix} i & j & k \\ 2 & 1 & -3 \\ 2 & -1 & 1 \end{vmatrix} = (-2, -8, -4).$$

(ii) An equation for π is: $-2(x + 1) - 8(y - 1) - 4(z - 2) = 0$ or $x + 4y + 2z - 7 = 0$.

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(C) all of the above

- (b) Let A be a nonzero 4×7 matrix, then A may be

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- (c) Let A be an $n \times n$ matrix in the reduced row echelon form. Then

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- (d) For all vectors U, V and W in $\mathbb{R}^n$

(A) $\|U + V + W\| \leq \|U\| + \|V\| + \|W\|$

- (e) Let V be a vector space and W_1 and W_2 be subspaces of V . If $B_1 = \{X, Y\}$ and $B_2 = \{X, Z\}$ are bases for W_1 and W_2 , respectively, then

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